EXPERIMENT 5: IMPLEMENTATION OF SUPPORT VECTOR MACHINE (SVM) FOR PULSAR STAR CLASSIFICATION WITH HYPERPARAMETER TUNING

1. DATASET SOURCE

The dataset used in this experiment is the **HTRU2 Pulsar Star Dataset**, which contains pulsar candidates collected during the High Time Resolution Universe Survey.

- Primary Source (UCI Repository): https://archive.ics.uci.edu/ml/datasets/HTRU2

2. DATASET DESCRIPTION

The HTRU2 dataset is designed to distinguish between **real pulsar signals** and **noise (non-pulsar signals)** using statistical features extracted from radio signals.

Dataset Characteristics:

- **Total Instances:** 17,898
- **Positive Class (Pulsar):** 1,639
- **Negative Class (Non-Pulsar):** 16,259
- **Number of Features:** 8 continuous features
- **Target Variable:** Binary (0 = Non-pulsar, 1 = Pulsar)
- **Data Type:** Numerical (floating-point values)
- **Class Imbalance:** Highly imbalanced dataset

Feature Description

Each instance represents a pulsar candidate and is described using 8 statistical features derived from signal processing:

1. Mean of the integrated profile
2. Standard deviation of the integrated profile
3. Excess kurtosis of the integrated profile
4. Skewness of the integrated profile
5. Mean of the DM-SNR curve
6. Standard deviation of the DM-SNR curve
7. Excess kurtosis of the DM-SNR curve
8. Skewness of the DM-SNR curve

3. MATHEMATICAL FORMULATION OF THE ALGORITHM

Support Vector Machine (SVM) is a **margin-based classifier** that finds an optimal decision boundary separating two classes.

Objective of SVM:

To maximize the margin between two classes while minimizing classification error.

Hard Margin SVM (Linearly Separable Case)

$$\min_{w,b} \frac{1}{2} || w ||^2$$

Subject to:

$$y_i(w \cdot x_i + b) \geq 1$$

Soft Margin SVM (Real-world Data)

$$\min_{w,b,\xi} \frac{1}{2} || w ||^2 + C \sum_{i=1}^{n} \xi_i$$

Where:

- ξ_i= slack variable (penalty for misclassification)
- C= regularization parameter

Constraint:

$$y_i(w \cdot x_i + b) \geq 1 - \xi_i$$

Kernel Trick (Non-linear Classification)

To handle non-linear data, SVM uses kernel functions:

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

Common Kernels Used:

- **Linear Kernel**
- **Polynomial Kernel**
- **Radial Basis Function (RBF):**

$$K(x_i, x_j) = \exp\left(-\gamma || x_i - x_j ||^2\right)$$

Interpretation:

- SVM transforms input space into higher dimensions
- Finds optimal hyperplane in transformed space
- Only **support vectors** influence the decision boundary

4. ALGORITHM LIMITATIONS

While SVM is powerful, it has several limitations, especially in the context of the pulsar dataset:

1. Sensitivity to Class Imbalance

- SVM may bias toward majority class (non-pulsars)
- Requires techniques like class weighting or resampling

2. Computational Complexity

- Training time increases with dataset size
- Kernel computations can be expensive

3. Kernel Selection Challenge

- Choosing the correct kernel is non-trivial
- Wrong kernel leads to poor performance

4. Hyperparameter Dependency

- Performance heavily depends on:
  - C
  - γ
- Requires extensive tuning

5. Lack of Interpretability

- Decision boundary is not easily interpreable
- Not ideal for explainable AI tasks

6. Sensitivity to Noise

- Outliers affect margin and support vectors
- Can degrade performance

5. METHODOLOGY / WORKFLOW

The experiment follows a structured pipeline:

Step 1: Data Loading

- Import dataset using pandas
- Inspect shape, types, and distributions

Step 2: Data Preprocessing

Handling Class Imbalance

- Techniques: Class weighting (class_weight='balanced')

Feature Scaling

- Apply **StandardScaler**
- Essential because SVM depends on distance calculations

Step 3: Train-Test Split

- Split dataset: 80% training and 20% testing

Step 4: Model Training

- Train SVM with: Linear Kernel / RBF Kernel

Step 5: Hyperparameter Tuning

- Use **GridSearchCV** and Cross-validation (e.g., 5-fold)

Step 6: Prediction and Evalaution

- Predict labels on test data and Compute performance metrics

6. PERFORMANCE ANALYSIS

Due to **class imbalance**, accuracy alone is not sufficient.

Evaluation Metrics Used:

1. Accuracy

$$Accuracy = \frac{TP + TN}{Total}$$

2. Precision

$$Precision = \frac{TP}{TP + FP}$$

3. Recall (Important for Pulsar Detection)

$$Recall = \frac{TP}{TP + FN}$$

4. F1 Score

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

7. HYPERPARAMETER TUNING

Hyperparameter tuning is crucial for optimizing SVM performance.

Parameters Tuned:

Parameter	Role
C	Controls margin vs misclassification
Gamma	Controls influence of points
Kernel	Determines decision boundary shape

Cross-Validation:

- 5-fold cross-validation used
- Prevents overfitting

Conclusion

The Support Vector Machine (SVM) proves to be an effective classifier for pulsar detection when combined with proper preprocessing and hyperparameter tuning. Given the highly imbalanced nature of the dataset, emphasis on recall and F1-score is crucial. The use of kernel functions, especially the RBF kernel, enables the model to capture non-linear patterns in the data. However, careful tuning and handling of imbalance are essential to achieve optimal performance.